

REVIEW ARTICLE

Cerebral Amyloid Angiopathy

Steven M. Greenberg, M.D., Ph.D.^{1,2}

SUMMARY

Cerebral amyloid angiopathy is a major cause of hemorrhagic stroke, a frequent contributor to age-related cognitive impairment, and a key component in adverse responses to beta-amyloid ($A\beta$) immunotherapy. Defined by pathological deposition of $A\beta$ in the small blood vessels of the brain, cerebral amyloid angiopathy is most often diagnosed on the basis of magnetic resonance imaging studies showing multiple hemorrhages or leptomeningeal blood products within or overlying the cerebral cortex. The disorder typically manifests as hemorrhagic stroke or as a contributing factor to cognitive decline and, less commonly, with transient focal neurologic symptoms or a cerebral inflammatory autoimmune syndrome. The high risk of recurrent hemorrhagic strokes associated with cerebral amyloid angiopathy poses a particular challenge in patients with indications for antithrombotic therapy and dictates a carefully individualized weighing of risks and benefits. Ongoing research is focused on tools to aid in risk prediction, early diagnostic markers, and identification of key pathogenic steps as targets for disease-modifying therapies.

Author affiliations are listed at the end of the article. Steven M. Greenberg can be contacted at sgreenberg@mgh.harvard.edu or at the Hemorrhagic Stroke Research Program, Massachusetts General Hospital Stroke Research Center, 175 Cambridge St., Suite 300, Boston, MA 02114.

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CME



CEREBRAL AMYLOID ANGIOPATHY IS AN AGE-ASSOCIATED CAUSE OF Cerebral hemorrhages, cognitive decline, and transient neurologic episodes that can mimic transient ischemic attacks. The disorder has hidden in plain sight through much of its history. In its common sporadic form, it is defined by deposition of the beta-amyloid ($A\beta$) peptide in walls of the small blood vessels of the brain and has a close relationship with the $A\beta$ -containing plaques of Alzheimer's disease. First reported as a postmortem observation by the German investigator Gustav Oppenheim in 1909¹ (just 2 years after Alois Alzheimer's published report of the disease that would eventually be named Alzheimer's disease), cerebral amyloid angiopathy was not linked to clinical events such as intracerebral hemorrhage until the second half of the 20th century. Subsequent progress has led to cerebral amyloid angiopathy being diagnosable in living patients. Along with its role in causing intracerebral hemorrhage and in contributing to age-related cognitive decline, cerebral amyloid angiopathy affects clinical decisions about the use of anticoagulation therapy in stroke treatment and prevention and the use of anti- $A\beta$ immunotherapies for Alzheimer's disease.

PATHOGENESIS AND EPIDEMIOLOGY

The $A\beta$ fibril deposits that characterize cerebral amyloid angiopathy differ from those in Alzheimer's disease in that they are primarily localized to cerebral blood-vessel walls rather than brain tissue and are characterized by a predominance of the shorter $A\beta$ 40 isoform rather than the longer $A\beta$ 42 isoform.² The pathological changes range from trace $A\beta$ deposits surrounding arteriolar smooth-

muscle cells (mild cerebral amyloid angiopathy, Fig. 1A) to full replacement of the arteriolar media (moderate cerebral amyloid angiopathy, Fig. 1A) and, ultimately, to fragmentation of the vessel wall (severe cerebral amyloid angiopathy, Fig. 1B).³ Amyloid is predominantly deposited in leptomeningeal and cortical arterioles, but it may appear in cerebral capillaries, otherwise sparing cerebral veins, venules, larger arteries, and deep penetrating arterioles in the basal ganglia, thalamus, and brain stem. Deposits are absent in vessels outside the central nervous system. Intracerebral hemorrhage occurs at the severe pathological stage³ and is likely to be the result of vascular remodeling that leads to replacement of A β -laden vessel-wall segments with fibrin and other plasma components, blood-brain barrier leakage, and accumulation of activated astrocytes and perivascular microglia or macrophages.⁴

Although the pathological changes that characterize cerebral amyloid angiopathy and Alzheimer's disease can appear independently, they co-occur more frequently than by chance. On the basis of pooled data from postmortem neuropathological studies, the estimated prevalence of moderate-to-severe cerebral amyloid angiopathy is 23.0% in the general population and 47.5% among patients with Alzheimer's disease.⁵ The tendency for the two diseases to overlap is probably related to shared pathophysiological mechanisms of A β deposition, as well as partially

shared predisposing risk factors, such as the presence and dose of the $\epsilon 4$ allele of APOE, the gene encoding apolipoprotein E.² The prevalence of moderate-to-severe cerebral amyloid angiopathy substantially exceeds the prevalence of hemorrhagic brain lesions related to cerebral amyloid angiopathy such as microbleeds (asymptomatic cerebral hemorrhages of <10 mm in diameter as detected on magnetic resonance imaging [MRI]), which have an estimated prevalence of 7.1% in the general population and 21.8% among persons with Alzheimer's disease.⁵ The fact that the

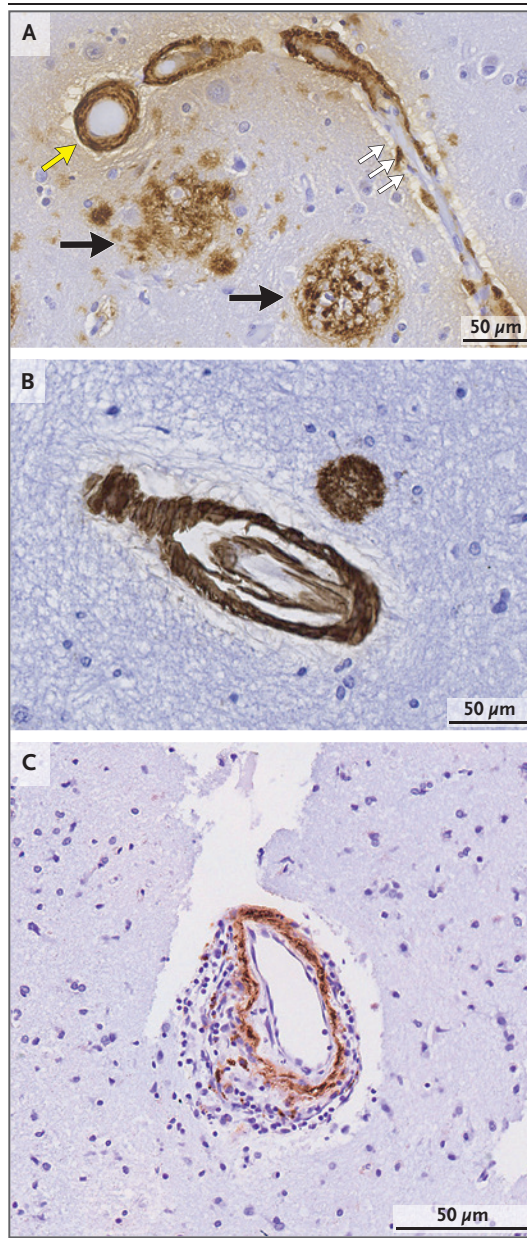


Figure 1. Neuropathological Features of Cerebral Amyloid Angiopathy.

The vessels in Panel A show both mild cerebral amyloid angiopathy, characterized by beta-amyloid (A β) deposits (white arrows) alternating with unaffected vessel segments, and moderate cerebral amyloid angiopathy, characterized by complete replacement of the vessel wall with A β (yellow arrow). Segment-to-segment variation in the severity of cerebral amyloid angiopathy within a brain is typical. The section also shows A β -containing senile plaques (black arrows). Panel B shows severe cerebral amyloid angiopathy, with concentric splitting of the A β -laden vessel wall creating a vessel-within-vessel appearance. The vessel segment in Panel C shows inflammation related to cerebral amyloid angiopathy, with an infiltration of mononuclear inflammatory cells surrounding an A β -positive segment. Anti-A β immunohistochemical staining was used for the sections in all three panels, with hematoxylin counterstaining. These neuropathological images are courtesy of Drs. Susanne van Veluw and Francesco Bax.

prevalence of cerebral amyloid angiopathy is higher than the prevalence of hemorrhagic lesions related to cerebral amyloid angiopathy supports the idea that brain hemorrhages develop in only a subgroup of persons with advanced cerebral amyloid angiopathy.

Much of our understanding of how cerebral amyloid angiopathy progresses from presymptomatic to symptomatic stages has come from longitudinal analysis of rare autosomal dominant, fully penetrant hereditary forms of cerebral amyloid angiopathy, such as that caused by the Dutch-type amyloid precursor protein (APP) mutation.⁶ Studies in APP mutation carriers suggest that the earliest vascular A β deposits occur approximately 30 to 40 years before the first intracerebral hemorrhage, a finding inferred from an observed reduction of A β in cerebrospinal fluid, a correlate of increased brain deposition, during that period. The next detectable change, occurring 20 to 30 years before the first intracerebral hemorrhage, is impaired cerebrovascular function, measured by functional MRI as reduced reactivity to physiologic stimulation, which probably reflects a loss of smooth vessel cells from arteriolar segments affected by cerebral amyloid angiopathy. The first evidence of brain-tissue injury, detected by sensitive measures of structural integrity such as diffusion-tensor MRI, appear 10 to 15 years before the first intracerebral hemorrhage. Finally, cerebral amyloid angiopathy-associated hemorrhagic lesions, a category that includes lobar intracerebral hemorrhage (i.e., a lesion situated within a lobe of the brain rather than in deep structures), microbleeds, and leptomeningeal foci of blood products (cortical superficial siderosis), appear as the culmination of this multidecade process.

Although this timeline is derived largely from studies involving APP mutation carriers, there is ancillary evidence of a similar decades-long timeline for the rare form of iatrogenic cerebral amyloid angiopathy appearing in persons who in early life had neurosurgical exposure to human cadaveric grafts containing A β . The mean interval between the presumed exposure and the first cerebral amyloid angiopathy-related symptomatic presentation in such persons is 36 years.⁷ The timeline for cerebral amyloid angiopathy appears to exceed the approximately 20-year interval between the first appearance of A β

biomarkers and the diagnosis of Alzheimer's disease,⁸ which suggests that cerebral amyloid angiopathy requires roughly one to two decades longer to progress from the first deposits of A β to symptomatic disease.

CLINICAL MANIFESTATIONS

Cerebral amyloid angiopathy can become apparent acutely as a symptomatic hemorrhagic stroke or insidiously as a contributor to progressive cognitive impairment. Intracerebral hemorrhage related to cerebral amyloid angiopathy, reflecting the locus of cerebral amyloid angiopathy, occurs predominantly in the superficial cortex, adjacent subarachnoid space, and subcortical white matter⁹ (Fig. 2A). Hemorrhages can also involve the superficial cerebellar cortex¹⁰ but spare deep gray-matter structures such as the putamen, thalamus, and brain stem, which are the regions most affected by arteriolosclerosis-related intracerebral hemorrhage. Symptoms of intracerebral hemorrhage related to cerebral amyloid angiopathy, like those of other hemorrhagic strokes, vary according to the location and volume of disrupted brain tissue.

In contrast to the acute stroke syndromes associated with intracerebral hemorrhage, cognitive decline associated with cerebral amyloid angiopathy evolves gradually over months and years without symptomatic stroke. The Rush Memory and Aging study and the Religious Orders study, which are longitudinal clinical-pathological studies, have shown that postmortem findings of moderate-to-severe cerebral amyloid angiopathy, as compared with mild or no cerebral amyloid angiopathy, are independently associated with worse cognitive performance and a faster rate of cognitive decline before death in an analysis adjusted for age, sex, educational level, and concomitant age-related neuropathological disease, including Alzheimer's disease, arteriolosclerosis, atherosclerosis, and other neurodegenerative processes.¹¹ This analysis estimated that when moderate-to-severe cerebral amyloid angiopathy disease is present (it was found in more than one third of autopsied brains in the study sample), it accounts for approximately 16% of a person's cognitive loss.

Cerebral amyloid angiopathy probably affects cognition through several types of brain lesions,

including MRI-detected white-matter hyperintensities¹² and cerebral microinfarcts.¹³ Like other vascular contributions to cognitive impairment and dementia, cerebral amyloid angiopathy has the greatest effect on processing speed and executive function.^{14,15} The clinical course of gradually worsening performance across multiple cognitive domains, including episodic memory, can be indistinguishable from the effects of Alzheimer's disease.

A third, increasingly recognized clinical manifestation of cerebral amyloid angiopathy is multiple brief spells (typically lasting less than 30

minutes) of focal motor weakness, sensory loss, tingling, dysarthria, or aphasia. Unlike the symptoms of transient ischemic attacks, the symptoms in these spells often spread along contiguous body parts over the course of minutes.¹⁶ These episodes frequently recur in clusters over a period of days or weeks in a stereotyped manner, with only slight variations in the timing and extent of spread. Computed tomography (CT) or MRI in patients with transient focal neurologic episodes due to cerebral amyloid angiopathy frequently discloses recent convexity subarachnoid hemorrhage or cortical superficial siderosis adjacent to cortical re-

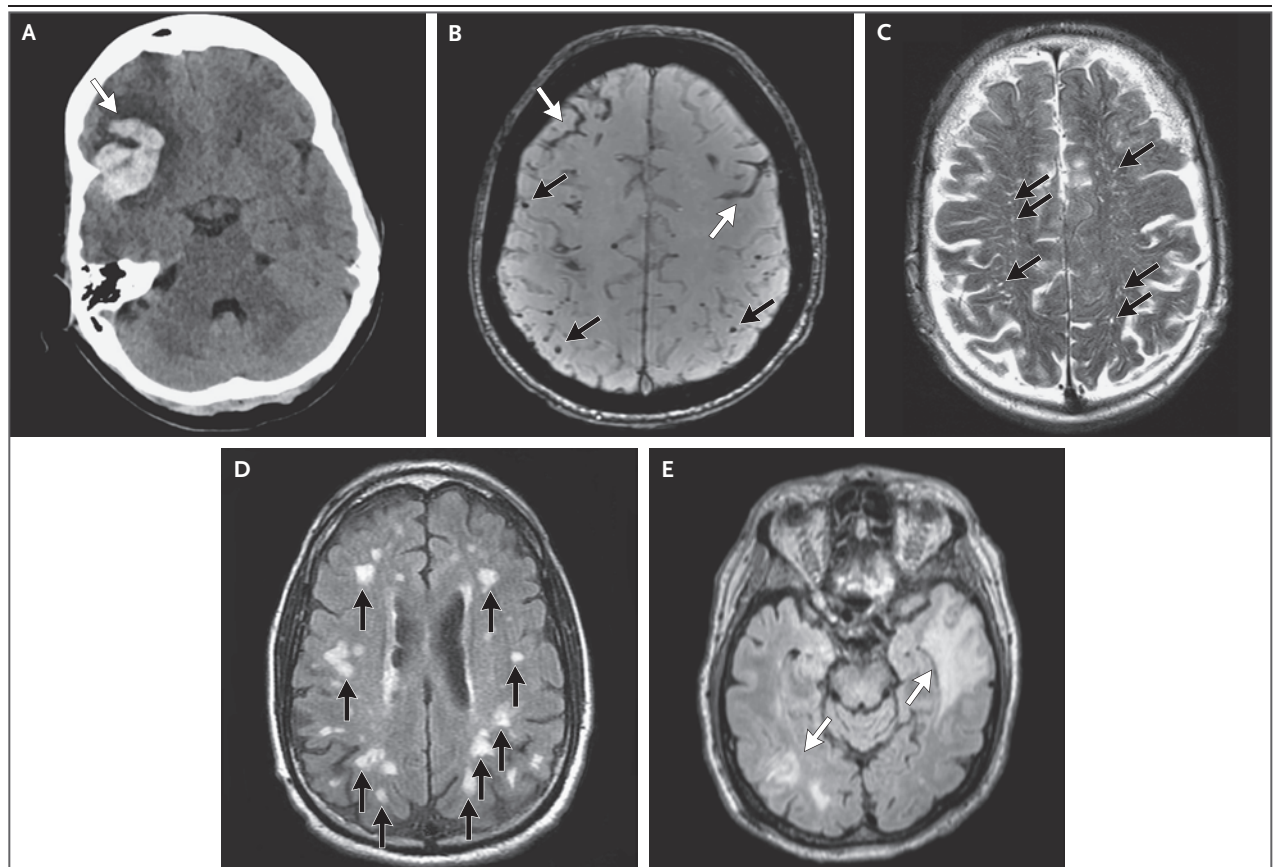


Figure 2. Neuroimaging Features of Cerebral Amyloid Angiopathy.

The computed tomographic scan in Panel A shows an acute right frontal lobar intracerebral hemorrhage, with subarachnoid extension and a fingerlike projection (arrow), features that are characteristic of cerebral amyloid angiopathy. In Panel B, magnetic resonance imaging with T2*-weighted images shows two foci of cortical superficial siderosis (white arrows) and multiple lobar cerebral microbleeds (black arrows). Panels C and D show the two types of nonhemorrhagic white-matter lesions incorporated into the Boston criteria for the diagnosis of cerebral amyloid angiopathy: severe (>20 in a hemisphere) visible perivascular spaces in the centrum semiovale seen on T2-weighted images (Panel C, arrows) and a multispot pattern of white-matter hyperintensities (>10 small, circular or ovoid lesions in the subcortical white matter of both hemispheres) on fluid-attenuated inversion recovery (FLAIR) images (Panel D, arrows). Panel E shows two FLAIR-hyperintense lesions in juxtacortical left frontal and right temporo-occipital white matter (arrows), which are characteristic of inflammation related to cerebral amyloid angiopathy, in a patient with a biopsy-confirmed diagnosis of this disorder.

gions associated with the symptoms. The finding of localized blood products and the spread of symptoms, two features that distinguish cerebral amyloid angiopathy–associated transient spells from typical transient ischemic attacks, are consistent with an underlying mechanism of cortical spreading depolarization, the same process that is thought to cause migraine auras.¹⁶

A fourth clinical manifestation, cerebral amyloid angiopathy–related inflammation, is discussed separately below because of its distinct clinical features, treatment, and implications for understanding amyloid-related imaging abnormalities (ARIA) associated with anti- $A\beta$ immunotherapy for Alzheimer's disease.

DIAGNOSIS

Hemorrhagic stroke, progressive cognitive decline, or transient focal neurologic episodes due to cerebral amyloid angiopathy may occur independently or in combination. Brain imaging in these situations often allows a diagnosis of cerebral amyloid angiopathy without the need to obtain brain tissue for pathological examination, either by hematoma appearance or most often by detection of multiple hemorrhages in characteristic locations. CT studies in patients with lobar intracerebral hemorrhage related to cerebral amyloid angiopathy tend to show blood in the subarachnoid space and fingerlike projections of the hematoma (Fig. 2A), findings that form the basis of the Edinburgh diagnostic criteria for cerebral amyloid angiopathy.⁹ In the study defining these criteria, the presence of both subarachnoid blood and fingerlike projections helped to rule in cerebral amyloid angiopathy (specificity, approximately 87%), and the absence of both findings helped to rule out cerebral amyloid angiopathy (sensitivity, 81%). The sensitivity of the Edinburgh criteria appears to be lower for intracerebral hemorrhages that are smaller than 15 ml.¹⁷

The Boston criteria for cerebral amyloid angiopathy (Table 1) are based on combinations of characteristic MRI findings, including lobar intracerebral hemorrhage, lobar microbleeds, and cortical superficial siderosis (Fig. 2B). These imaging findings are accurately detected by T2*-weighted MRI sequences such as gradient-echo and susceptibility-weighted imaging. The pres-

ence of at least two lobar hemorrhagic lesions (intracerebral hemorrhage, microbleeds, or foci of cortical superficial siderosis) without hemorrhages in deep-brain regions or other causes of hemorrhage meets the Boston criteria for probable cerebral amyloid angiopathy (Table 1). The most recent revision of the Boston criteria also allows the diagnosis of probable cerebral amyloid angiopathy in the presence of one lobar hemorrhagic lesion plus one nonhemorrhagic MRI marker associated with cerebral amyloid angiopathy: severe visible perivascular spaces in the centrum semiovale (Fig. 2C) or white-matter hyperintensities in a multispot pattern (Fig. 2D).

In an MRI–pathological validation analysis, the imaging diagnosis of probable cerebral amyloid angiopathy according to the Boston criteria corresponded to pathological evidence of moderate-to-severe cerebral amyloid angiopathy in the brain at autopsy with approximately 74.5% sensitivity and 95.0% specificity,¹⁸ with the greatest sensitivity (90.2%) for patients presenting with lobar intracerebral hemorrhage and lower sensitivity (55.1%) for those with clinical presentations such as cognitive decline in the absence of intracerebral hemorrhage. The higher diagnostic sensitivity with clinical presentations that include intracerebral hemorrhage supports the idea that cerebral amyloid angiopathy can be advanced enough to contribute to cognitive impairment without necessarily triggering even the one hemorrhagic lesion that the Boston criteria require for the diagnosis of probable cerebral amyloid angiopathy (Table 1). The category of probable cerebral amyloid angiopathy with supporting pathological findings applies when cerebral amyloid angiopathy–positive tissue is available from a brain biopsy or from tissue obtained during hematoma evacuation. The diagnosis of definite cerebral amyloid angiopathy under these criteria requires evidence of severe cerebral amyloid angiopathy in the brain at autopsy.

Positron-emission tomography with amyloid-selective tracers has been shown to detect cerebral amyloid angiopathy¹⁹ but does not readily distinguish between Alzheimer's disease and cerebral amyloid angiopathy. Cerebrospinal fluid from patients with cerebral amyloid angiopathy shows reductions in both the $A\beta$ 42 isoform and the $A\beta$ 40 isoform,²⁰ findings that can help con-

firm the diagnosis and may help differentiate cerebral amyloid angiopathy from Alzheimer's disease (which does not cause a reduced level of the A β 40 isoform in cerebrospinal fluid).

TREATMENT AND PROGNOSIS

Other than its characteristic lobar location, acute intracerebral hemorrhage related to cerebral amyloid angiopathy does not systematically differ from other forms of acute intracerebral hemorrhage with respect to clinical appearance, course, or response to treatment. The risk of hematoma expansion, for example, did not differ significantly between 202 patients with lobar intracerebral hemorrhage that met the Edinburgh or Boston criteria for cerebral amyloid angiopathy and 1637 patients with nonlobar intracerebral hemorrhage who were enrolled in the Tranexamic Acid for Hyperacute Primary Intracerebral Haemorrhage trial.²¹ With regard to the safety of hematoma evacuation, a systematic review of 352 suspected cases of intracerebral hemorrhage related to cerebral amyloid angiopathy in which patients underwent surgical intervention (primarily open craniotomy) showed a low incidence of intraoperative hemorrhage (0.6%, 2 of 352 patients), and the estimated incidence of postoperative hemorrhage was similar to that for other types of intracerebral hemorrhage.²² Current guidelines regarding the diagnosis and treatment of acute intracerebral hemorrhage, as well as the prognosis, do not draw distinctions between intracerebral hemorrhages that are related and those that are unrelated to cerebral amyloid angiopathy.^{23,24}

Nevertheless, the recurrence rate is higher for hemorrhage associated with cerebral amyloid angiopathy (typically at other lobar locations in the brain) than for other forms of intracerebral hemorrhage. A meta-analysis of data from 1306 patients with intracerebral hemorrhage showed that the annual rate of recurrence was 7.4% (95% confidence interval [CI], 3.2 to 12.6) for intracerebral hemorrhage that was related to cerebral amyloid angiopathy as compared with 1.1% (95% CI, 0.5 to 1.7) for intracerebral hemorrhage that was unrelated to cerebral amyloid angiopathy,²⁵ which places cerebral amyloid angiopathy-related intracerebral hemorrhage among the most frequently recurrent stroke subtypes. Re-

Table 1. Boston Criteria, Version 2.0, for Probable Cerebral Amyloid Angiopathy.*

Age $\geq$ 50 years
Clinical presentation with spontaneous intracerebral hemorrhage, cognitive decline, or transient focal neurologic symptoms
One of the following: - Two or more of the following lobar hemorrhagic lesions on T2*-weighted MRI: intracerebral hemorrhages, cerebral microbleeds, foci of cortical superficial siderosis, or convexity subarachnoid hemorrhage - One of the above lobar hemorrhagic lesions plus either of the following nonhemorrhagic white-matter features: severe perivascular spaces in the centrum semiovale or white-matter hyperintensities in a multispot pattern
Absence of spontaneous hemorrhagic lesions on T2*-weighted MRI in deep-brain regions atypical of cerebral amyloid angiopathy: basal ganglia, thalamus, and brain stem
Absence of other direct causes of intracranial hemorrhage: severe head trauma, hemorrhagic transformation of ischemic strokes, arteriovenous malformation, hemorrhagic tumor, central nervous system vasculitis, and aneurysmal subarachnoid hemorrhage

* The Boston criteria, version 2.0, were derived from a cohort of 159 patients with intracerebral hemorrhage, cognitive impairment, or transient focal neurologic episodes who underwent MRI and brain-tissue biopsy for histopathological evaluation, with validation in two independent cohorts of 59 and 123 patients.¹⁸ Information in this table is adapted from Charidimou et al.¹⁸

currence rates have been highest among patients with cortical superficial siderosis (particularly multifocal or disseminated siderosis) or multiple previous intracerebral hemorrhages.²⁵⁻²⁷ Conversely, patients in whom cerebral amyloid angiopathy is diagnosed on the basis of multiple lobar microbleeds alone, in the absence of larger intracerebral hemorrhages and cortical superficial siderosis, appear to be at relatively low risk for future intracerebral hemorrhage, with an estimated risk of 6% over a period of 5 years in a single-center study involving 80 probable cerebral amyloid angiopathy cases.²⁸ Other factors reported to be associated with an increased risk of recurrent intracerebral hemorrhage related to cerebral amyloid angiopathy are higher blood pressures²⁹ and an APOE genotype containing the ϵ 4 or ϵ 2 allele.³⁰ Although the APOE genotype affects the risk of recurrent hemorrhage, it is typically not assayed in the care of patients with cerebral amyloid angiopathy.

The risk of a future intracerebral hemorrhage is an important consideration in decisions about antithrombotic therapy for patients with cerebral amyloid angiopathy. For the relatively common situation of nonvalvular atrial fibrillation

KEY POINTS

CEREBRAL AMYLOID ANGIOPATHY

- Cerebral amyloid angiopathy, the deposition of beta-amyloid peptide in small blood vessels of the brain, is a common age-associated disease that, when advanced, can cause vessel breakdown and bleeding.
- The major clinical manifestations of cerebral amyloid angiopathy are hemorrhagic strokes that frequently recur and smaller brain injuries that contribute to progressive cognitive decline.
- Cerebral amyloid angiopathy can be diagnosed with high specificity on the basis of magnetic resonance imaging showing multiple bleeds or microbleeds confined to the cortex or leptomeninges, as well as other, nonhemorrhagic lesions in the white matter.
- Decisions regarding antithrombotic treatment for patients with both cerebral amyloid angiopathy and a defined indication for anticoagulation require an individualized balancing of risks and benefits, which includes consideration of the patient's risk of future hemorrhagic strokes.
- Cerebral amyloid angiopathy can trigger a cerebral autoimmune inflammatory syndrome, a condition mimicking the amyloid-related imaging abnormalities identified as the major adverse effect of Alzheimer's disease immunotherapy.
- The priorities for improving prevention of cerebral amyloid angiopathy–related hemorrhage are the development of methods for early detection and candidate treatments for blocking the cascade of pathogenic steps.

in patients with probable cerebral amyloid angiopathy, multiple completed and ongoing randomized clinical trials have not provided clarity about whether to provide anticoagulation therapy.^{31,32} Clinicians are therefore advised to weigh the predictors of cardioembolic stroke related to nonvalvular atrial fibrillation³³ against the predictors of intracerebral hemorrhage related to cerebral amyloid angiopathy (listed above) on an individual basis. After consideration of the balance between these factors along with the poor outcomes among patients who have an intracerebral hemorrhage during anticoagulation therapy,³⁴ the decision might be to administer a direct-acting oral anticoagulant, perform percutaneous left atrial appendage occlusion,³⁵ or administer antiplatelet monotherapy. For example, my approach to a patient with cerebral amyloid angiopathy who has atrial fibrillation–related stroke and microbleeds but no intracerebral hemorrhage or cortical superficial siderosis might be to consider using a direct oral anticoagulant as a preventive agent, whereas for a patient with cerebral amyloid angiopathy, low-risk atrial fibrillation, and multiple previous intracerebral hemorrhages, I might offer enrollment in a clinical trial or alternatives to anticoagulation, such as left atrial appendage occlusion or antiplatelet monotherapy, which did not appear to substantially increase the risk of recurrent intracerebral hemorrhage in a randomized trial.³⁶ The

vitamin K antagonist warfarin is typically avoided in patients with cerebral amyloid angiopathy in situations in which direct oral anticoagulants are effective because of the higher risk of intracerebral hemorrhage with warfarin.³⁷

Blood pressure control (target pressure, <130/80 mm Hg) is recommended for secondary prevention of intracerebral hemorrhage,^{23,24} with data suggesting that the benefits also apply to cerebral amyloid angiopathy.^{29,38} On the basis of epidemiologic data, avoidance of heavy alcohol use is reasonable for patients who have had an intracerebral hemorrhage.³⁹ Evidence is equivocal on whether other medication classes, such as statins⁴⁰ or selective serotonin-reuptake inhibitors,⁴¹ increase the risk of recurrent intracerebral hemorrhage; an individualized approach in which benefits for defined indications are balanced against risks that have not yet been convincingly demonstrated may therefore be warranted. Although healthy lifestyle and dietary choices are reasonable for all types of cerebrovascular disease, evidence that clearly links lifestyle or dietary factors to cerebral amyloid angiopathy is currently lacking.

CEREBRAL AMYLOID ANGIOPATHY–RELATED INFLAMMATION AND AMYLOID-RELATED IMAGING ABNORMALITIES (ARIA)

The syndrome of inflammation related to cerebral amyloid angiopathy appears to represent an autoimmune response directed to the cerebro-

vascular A β deposits (Fig. 1C)^{42,43}; the response may be driven by spontaneous generation of anti-A β autoantibodies.⁴⁴ Characteristic symptoms of this distinctive form of cerebral amyloid angiopathy are headache, seizures, personality changes, encephalopathy, and focal neurologic deficits, typically emerging over a period of weeks to months.⁴⁵ In the absence of neuropathological confirmation,⁴² a diagnosis of probable cerebral amyloid angiopathy–related inflammation⁴⁶ can be established on the basis of MRI studies showing single or multiple patches of white-matter hyperintensities involving juxtacortical white matter (Fig. 2E) in patients with hemorrhagic lesions characteristic of probable cerebral amyloid angiopathy. Observational studies suggest that a course of immunosuppressive therapy such as high-dose glucocorticoids can result in clinical and imaging improvement in patients with cerebral amyloid angiopathy–related inflammation⁴⁷ and may reduce the likelihood of recurrent disease flares,⁴⁸ which makes this an important syndrome to identify.

An intriguing feature of cerebral amyloid angiopathy–related inflammation is its close resemblance to ARIA — an important adverse effect of anti-A β immunotherapy for Alzheimer's disease⁴⁹ — in its clinical presentation, appearance on neuroimaging, and neuropathological features. It is plausible that ARIA represents an iatrogenic form of spontaneously occurring inflammation related to cerebral amyloid angiopathy, a hypothesis supported by the observation that patients with microbleeds or cortical superficial siderosis who undergo immunotherapy have an increased risk of ARIA.⁵⁰ Although these data implicate cerebral amyloid angiopathy as a risk factor for ARIA, current guidelines support the use of anti-A β immunotherapy in patients with Alzheimer's disease who have had up to four microbleeds, after a discussion of the potential risks and benefits.^{51,52}

CONCLUSIONS AND FUTURE DIRECTIONS

A neuroimaging-based diagnosis of cerebral amyloid angiopathy can inform decisions about blood-pressure control and judicious use or avoidance of antithrombotic therapies. Similar considerations of risks and benefits apply to the incorporation of cerebral amyloid angiopathy markers into decisions about whether to initiate immunotherapy in patients with Alzheimer's disease.

The limited treatment options for cerebral amyloid angiopathy highlight the importance of identifying new approaches to disease modification. The decades-long pathogenic pathway and insidious effects on physiological properties of blood vessel, cognitive function, and intracerebral hemorrhage risk provide a long time window for slowing progression, but only if the early pathogenic steps can be detected and blocked. Potential approaches to slowing the progression of cerebral amyloid angiopathy include reducing A β production, enhancing A β clearance, and protecting vessels from destructive remodeling triggered by A β . Data on blocking production of A β may emerge from an ongoing phase 2 trial (ClinicalTrials.gov number, NCT06393712) of a C16-conjugated short interfering RNA⁵³ targeted to APP, a trial involving persons with symptomatic sporadic cerebral amyloid angiopathy as well as those with presymptomatic or symptomatic Dutch-type cerebral amyloid angiopathy. Determining how and when cerebral amyloid angiopathy can be slowed will inform future progress in developing treatments for this small-vessel brain disease.

Disclosure forms provided by the author are available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

¹J. Philip Kistler Stroke Research Center, Department of Neurology, Massachusetts General Hospital, Boston; ²Harvard Medical School, Boston.

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